

The Surplus From a Sale and Who Captures It

**ECON 201: Principles of Microeconomics
Lecture 10**

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Last Class: Choosing Price and Quantity

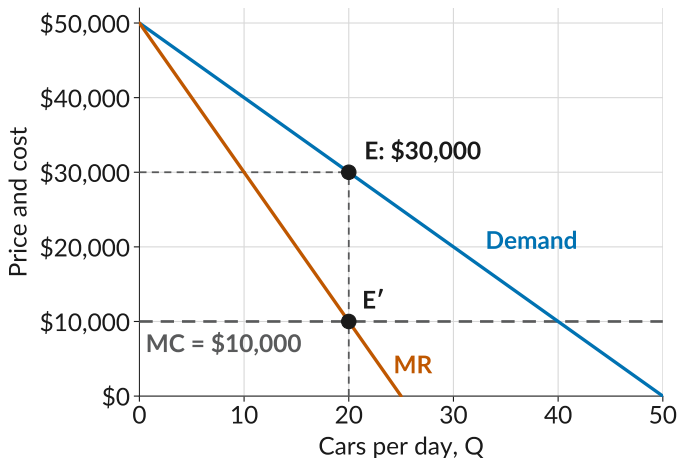
- **Marginal revenue** (MR) is the increase in revenue when one more unit is sold. To sell it, the firm lowers the price on every unit, so MR is below the price.
- **Marginal cost** (MC) is the increase in cost when one more unit is produced.
- While MR is above MC, one more unit adds to profit. Once MR falls below MC, it lowers profit. So the firm produces where $MR = MC$.
- The demand curve then gives the price it can charge for that quantity.

Today: when something is bought and sold, how much does everyone gain from the sale?

Beautiful Cars

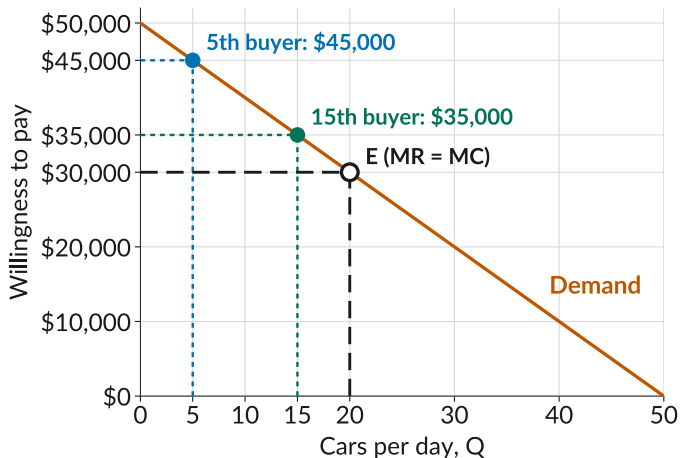
- Each car costs Beautiful Cars \$10,000 to make.
- MR equals MC at 20 cars, at E' .
- At 20 cars, the demand curve gives a price of \$30,000, at E .

Who gains from these sales, and by how much?



Demand as Willingness to Pay

- The height of the demand curve at Q is the willingness to pay (WTP) of the Q th buyer.
- Consider the 5th buyer. What is the total gain from the car sold to this buyer at a price of \$30,000?



The Surplus on One Car

The 5th buyer's willingness to pay for a car is \$45,000. What is the **buyer's gain** from the sale?

$$\text{Buyer's gain} = \text{WTP} - P = 45,000 - 30,000 = 15,000$$

What is the **firm's gain** from the sale? It is the profit on this one car:

$$\text{Firm's gain} = P - \text{MC} = 30,000 - 10,000 = 20,000$$

The **joint surplus** (a measure of the gains from trade) is the sum of the economic rents of all involved in an economic interaction:

$$\text{Joint surplus} = \underbrace{(\text{WTP} - P)}_{\text{buyer's gain}} + \underbrace{(P - \text{MC})}_{\text{firm's gain}} = \text{WTP} - \text{MC} = 35,000$$

The Surplus on Two Cars

Doing the same for the 15th buyer, whose willingness to pay is \$35,000:

Buyer	WTP	Joint surplus, $WTP - MC$	Buyer's gain, $WTP - P$	Firm's gain, $P - MC$
5th	45,000	35,000	15,000	20,000
15th	35,000	25,000	5,000	20,000

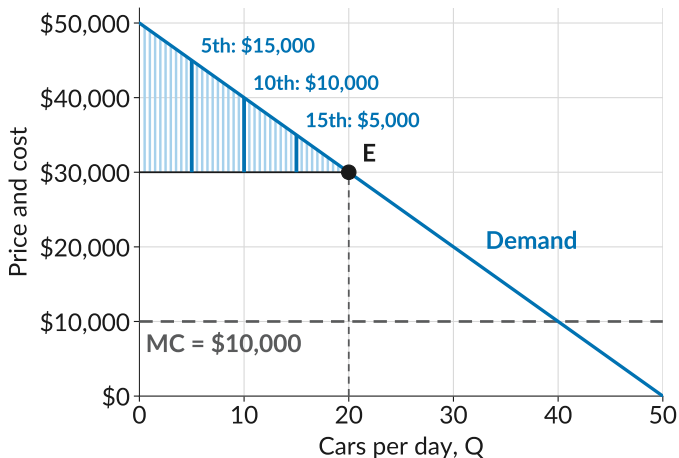
- Every buyer pays the same price, so the firm's gain is \$20,000 on every car.
- The buyer's gain shrinks as willingness to pay falls.

What is the total gain in the market, adding up the gains of all the buyers and the firm's gains on all the cars it sells?

Consumer Surplus

- Each consumer's surplus from a sale is $WTP - P$.
- Consumer surplus (CS):** the sum of these surpluses across all consumers.
- Calculate using the area of a triangle:

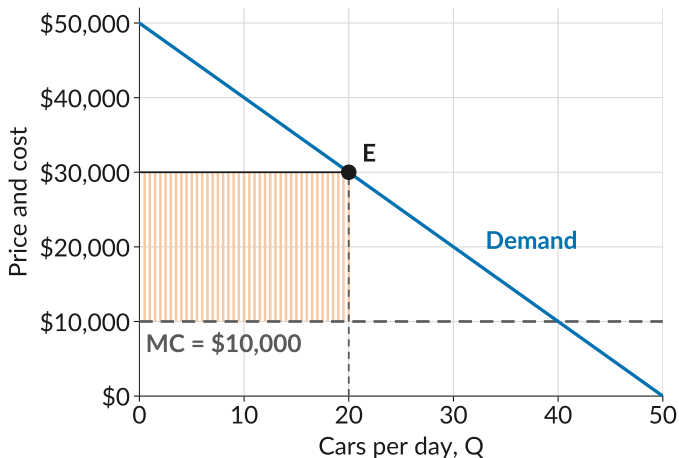
$$\begin{aligned} CS &= \frac{1}{2} \times \text{base} \times \text{height} \\ &= \frac{1}{2} \times 20 \times 20,000 \\ &= 200,000 \end{aligned}$$



Producer Surplus

- The firm's surplus from each sale is $P - MC$.
- **Producer surplus (PS)**: the sum of these surpluses across all units sold.
- Calculate using the area of a rectangle:

$$\begin{aligned} \text{PS} &= \text{base} \times \text{height} \\ &= 20 \times 20,000 \\ &= 400,000 \end{aligned}$$



Producer Surplus and Profit

Recall that Beautiful Cars' cost function is $C(Q) = 60,000 + 10,000Q$, with a fixed cost of \$60,000 a day. Its profit from 20 cars at \$30,000 is

$$\text{Profit} = \underbrace{30,000 \times 20}_{\text{Revenue}} - \underbrace{(60,000 + 10,000 \times 20)}_{\text{Cost}} = 340,000$$

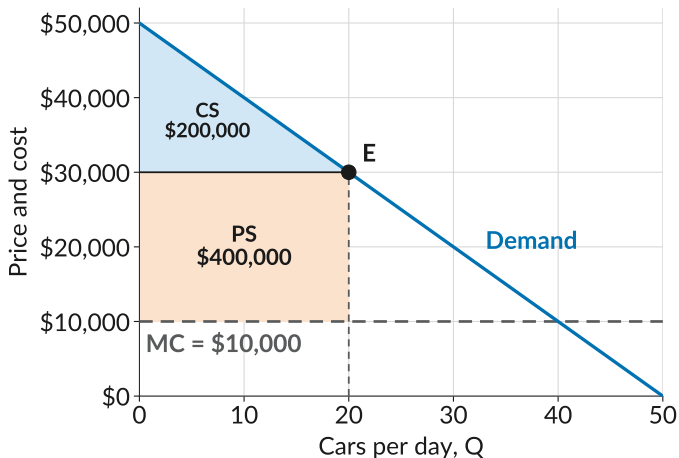
This is less than the producer surplus of \$400,000. With a fixed cost F , profit is

$$\text{Profit} = P \times Q - (\underbrace{MC \times Q}_{\text{Producer Surplus}} + \underbrace{F}_{\text{Fixed Cost}}) = \underbrace{(P - MC) \times Q}_{\text{Producer Surplus}} - \underbrace{F}_{\text{Fixed Cost}}$$

Producer surplus leaves out the fixed cost because it counts only what each sale adds. Selling one more car, or one fewer, does not change the \$60,000 the factory costs.

Is This a Good Outcome?

- The 20 sales create \$600,000 of surplus a day.
- How do we judge if this is a good market outcome?
- One criterion economists use is **Pareto efficiency**.



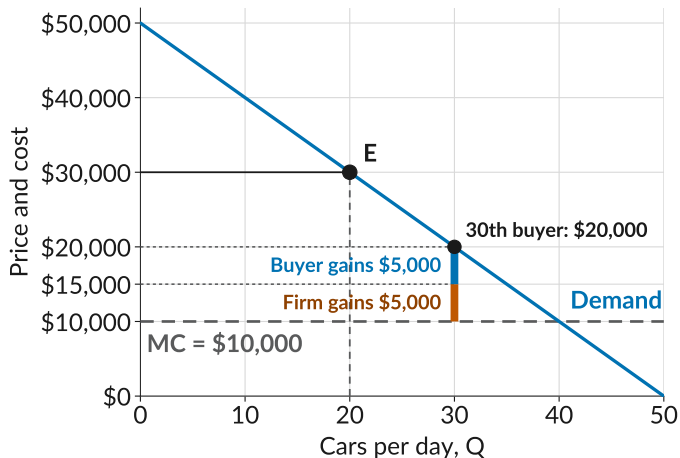
Pareto Efficiency

- An outcome is **Pareto efficient** if there is no other (feasible) outcome in which at least one person would be better off, and nobody worse off.
- We say there is a possible **Pareto improvement** if some change would make at least one person better off without making anyone worse off.
- Example: say you order pizza with your friends, and one friend eats all of it. Is that Pareto efficient?

In the market for Beautiful Cars, could there be a Pareto improvement? Is there a buyer who values a car at more than the \$10,000 it costs to make, but does not get one?

Buyers Left Out

- The 30th buyer's WTP is \$20,000, above the \$10,000 MC.
- At \$30,000, this buyer does not buy.
- Sell this buyer one car at \$15,000: each side gains \$5,000.
- Nobody is worse off, so this is a Pareto improvement.



Why Does Beautiful Cars Stop at 20?

In fact, buyers 21 to 40 all value a car above its \$10,000 MC but do not get one, so the outcome at E is not Pareto efficient. What is stopping Beautiful Cars from selling to them?

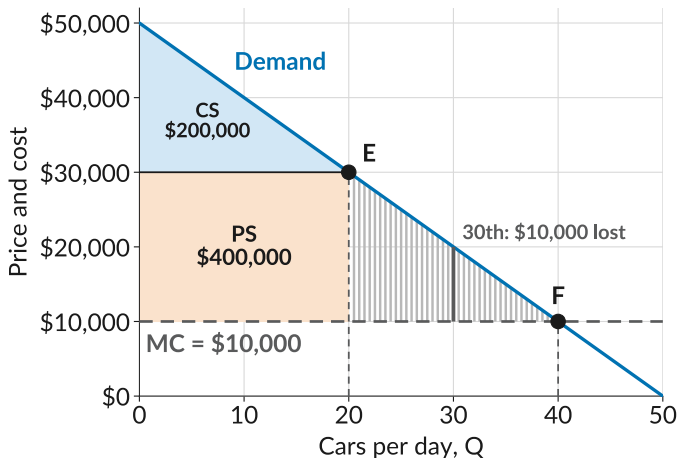
- It sets a single price for all buyers. Selling to each buyer at a different price is hard to arrange, and it is hard to know each buyer's willingness to pay.
- To reach the 30th buyer, it would have to charge \$20,000 to all 30 buyers, which would lower its profit.
- Setting different prices for different buyers, based on their willingness to pay, is called **price discrimination**. Where do we see this around us?

Can we measure the surplus lost from these missing sales?

Deadweight Loss

- Buyers 21 to 40 value a car above its MC but do not get one, so each loses $WTP - MC$.
- **Deadweight loss (DWL):** the total surplus lost.
- Calculate using the area of a triangle:

$$\begin{aligned} DWL &= \frac{1}{2} \times 20 \times 20,000 \\ &= 200,000 \end{aligned}$$



Who Captures the Surplus?

The joint surplus from a sale does not depend on the price:

$$\text{Joint surplus} = \text{WTP} - \text{MC}$$

- The price P decides the split: the buyer gets $\text{WTP} - P$ and the firm gets $P - \text{MC}$.
- The split depends on **bargaining power**: each side's ability to push the price in its favor.
- Beautiful Cars is the only seller of its car, so it has **market power**: it can set a high price, and buyers who value the car highly will still pay it.
- One buyer cannot bargain for a better deal, because the firm has many other customers. The firm takes \$400,000 of the \$600,000.

The Deadweight Loss of Christmas

- Economist Joel Waldfogel calculated the deadweight loss of Christmas.
- He asked college students what each gift cost, and what they would have paid for it.
- Gifts were worth 10% to a third less than they cost: a \$4 to \$13 billion deadweight loss in the US in 1992.
- A dollar from friends was worth about 99 cents; from grandparents, about 63 cents (though grandparents seem to know it: 43% of their gifts were cash).

GIFT RECEIPT

From: Grandma

1 ugly sweater

Price paid	\$50.00
Value to you	\$31.50

**Deadweight
loss** **\$18.50**

No returns. No exchanges.
Thank you for your love.

Source: Waldfogel (1993), *American Economic Review* 83(5).

What to Do Next

- Before next class: read section 7.7, and go over the slides and the worksheet.
- Section 7.7 also asks what happens if the firm could charge every buyer their full willingness to pay (Exercise 7.3). We skip that exercise for now.
- Work through the practice problems for this lecture.
- Next class we look at how firms stand out from the crowd, and how competition limits the prices they can set.